# E-commerce Sales

## Exploratory Data Analysis Report

| | |
|---|---|
| **Report Type:** | Exploratory Data Analysis (EDA) |
| **Dataset:** | E-commerce Sales Data |
| **Total Records:** | 982 transactions |
| **Analysis Date:** | 2024 |
| **Tools Used:** | Python, Pandas, Matplotlib, Seaborn |

# 1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis of e-commerce sales data containing 982 transactions. The analysis covers customer demographics, product preferences, geographic distribution, temporal patterns, and shipping performance. Key findings: Electronics dominates sales revenue, Laptop is the top product, East region generates highest revenue, and November 2023 was the peak sales month. The return rate of 30.4% requires immediate attention.

# 2. Introduction

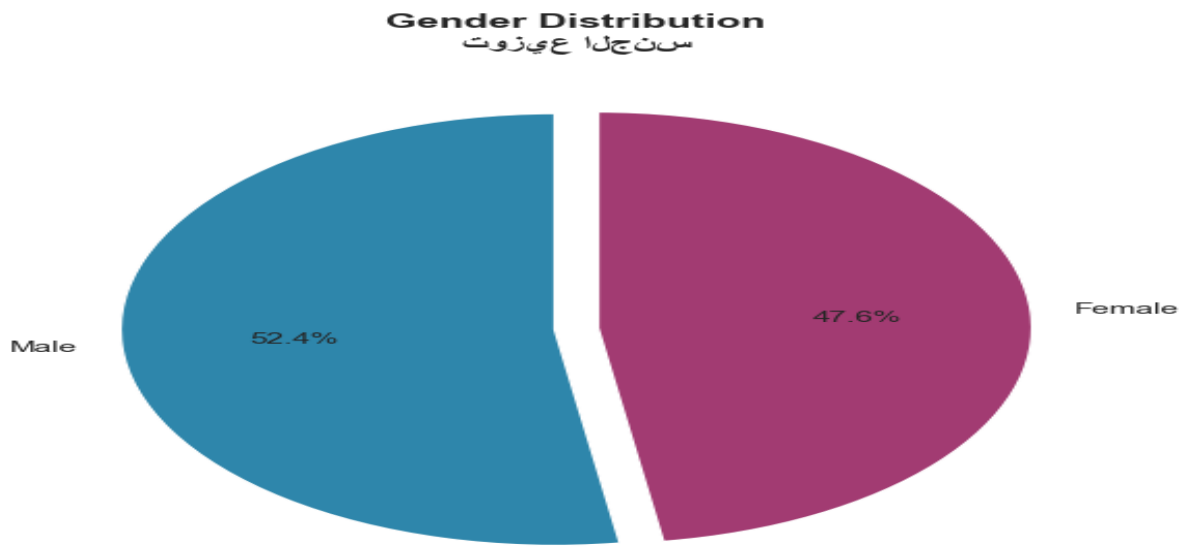
Exploratory Data Analysis (EDA) is a critical first step in any data science project. This project analyzes real e-commerce sales data to answer key business questions: What are the most popular products? Which regions generate the most revenue? How do customer demographics affect purchasing behavior? What are the temporal patterns?

# 3. Visual Analysis

This section presents all visualizations organized by analysis type.

## Gender Distribution

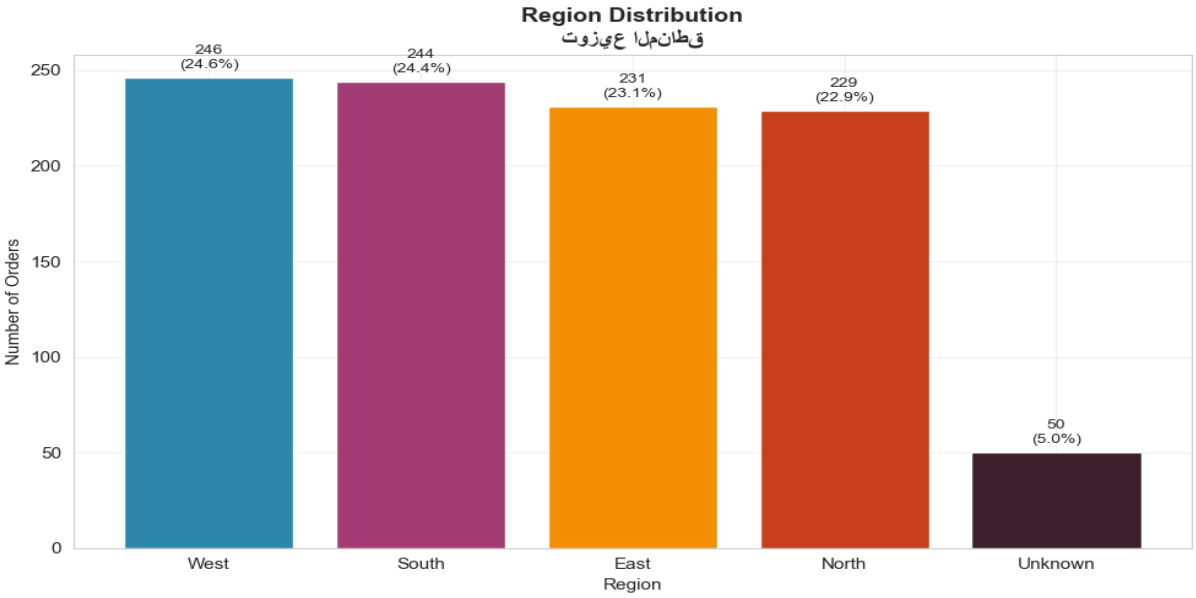
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## Region Distribution

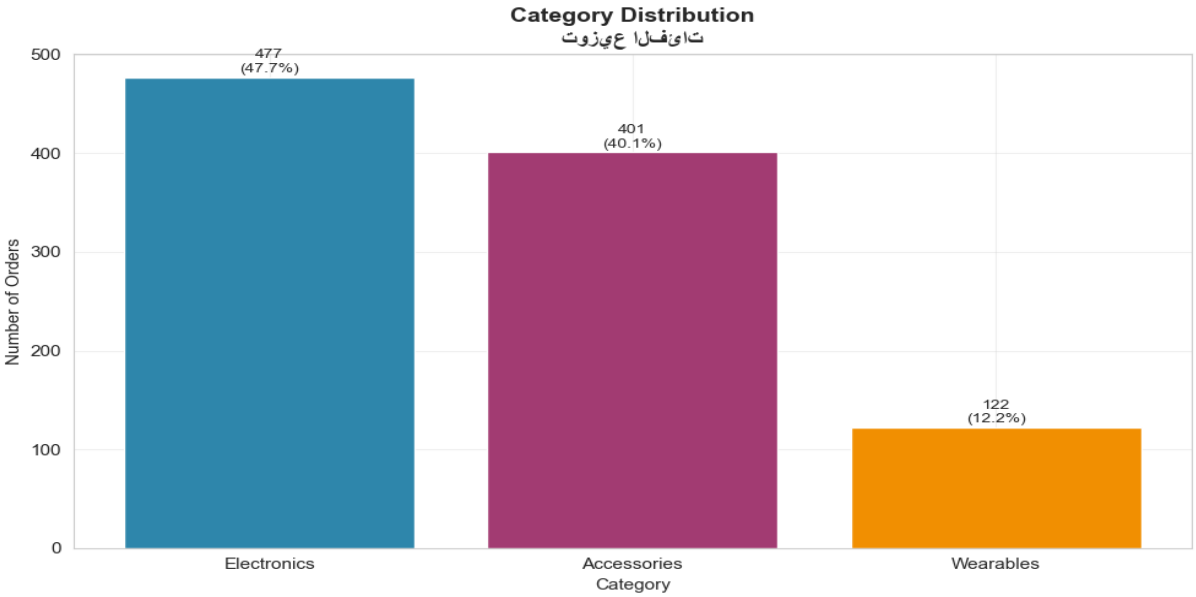
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## Category Distribution

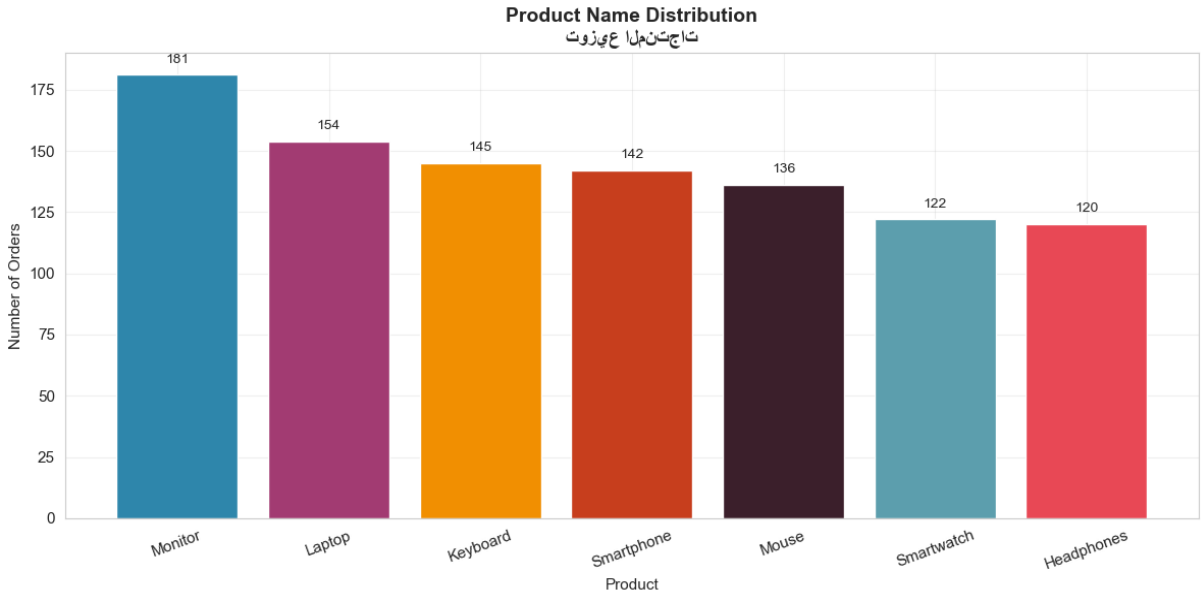
[illegible] [illegible]




[illegible] [illegible] [illegible] [illegible] [illegible]. Electronics [illegible] [illegible] [illegible].

## Product Distribution

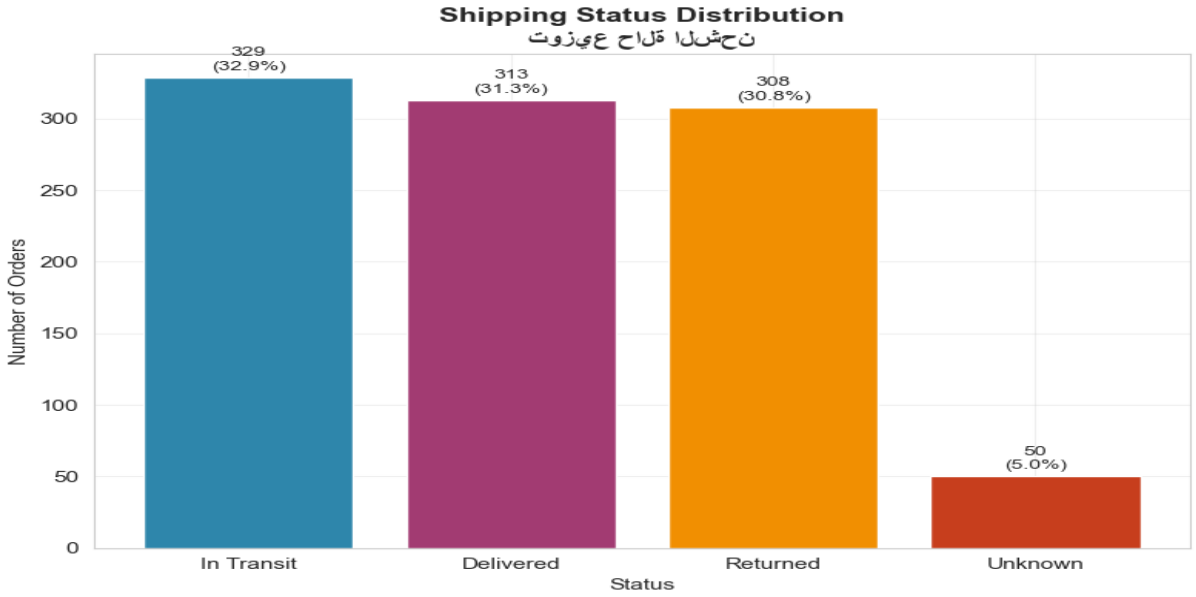
[illegible] [illegible]




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## Shipping Status

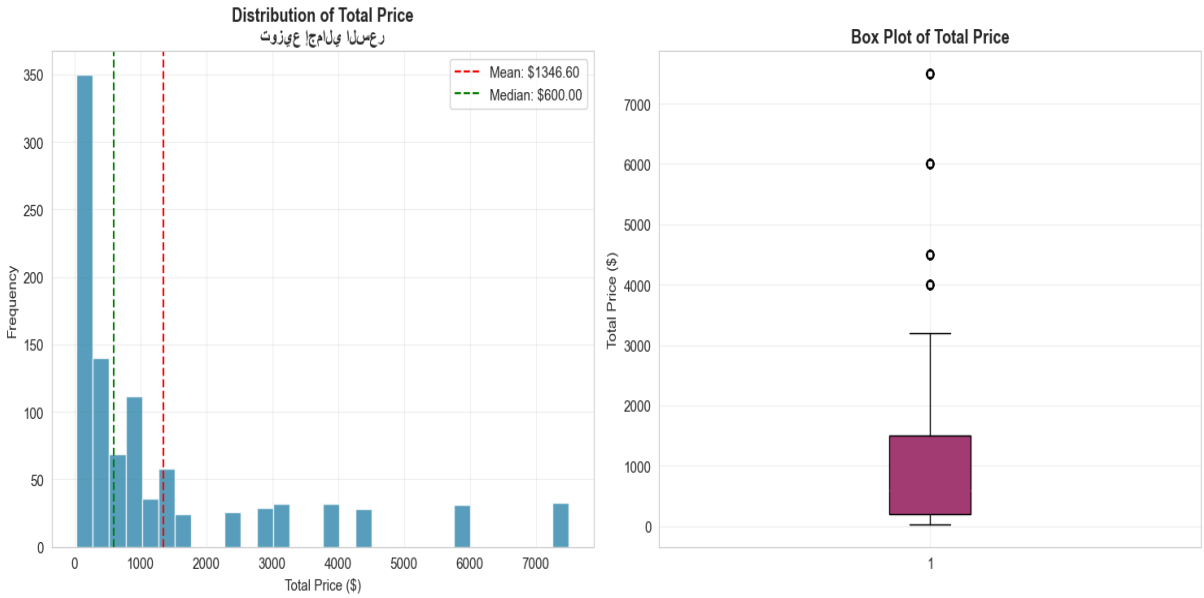
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## Price Distribution

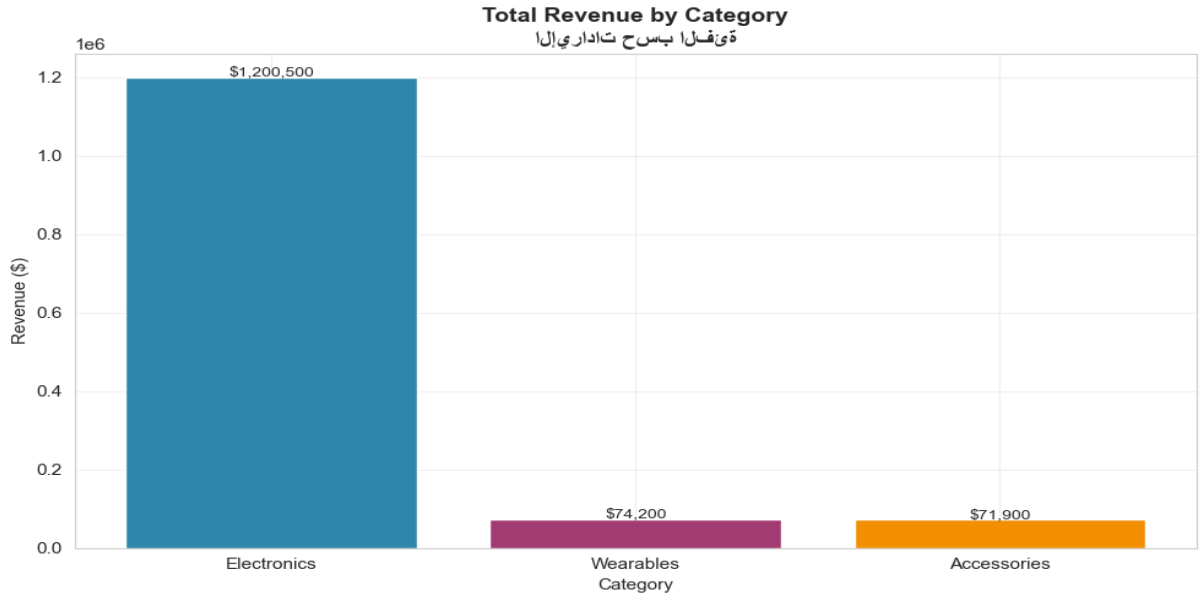
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## Revenue by Category

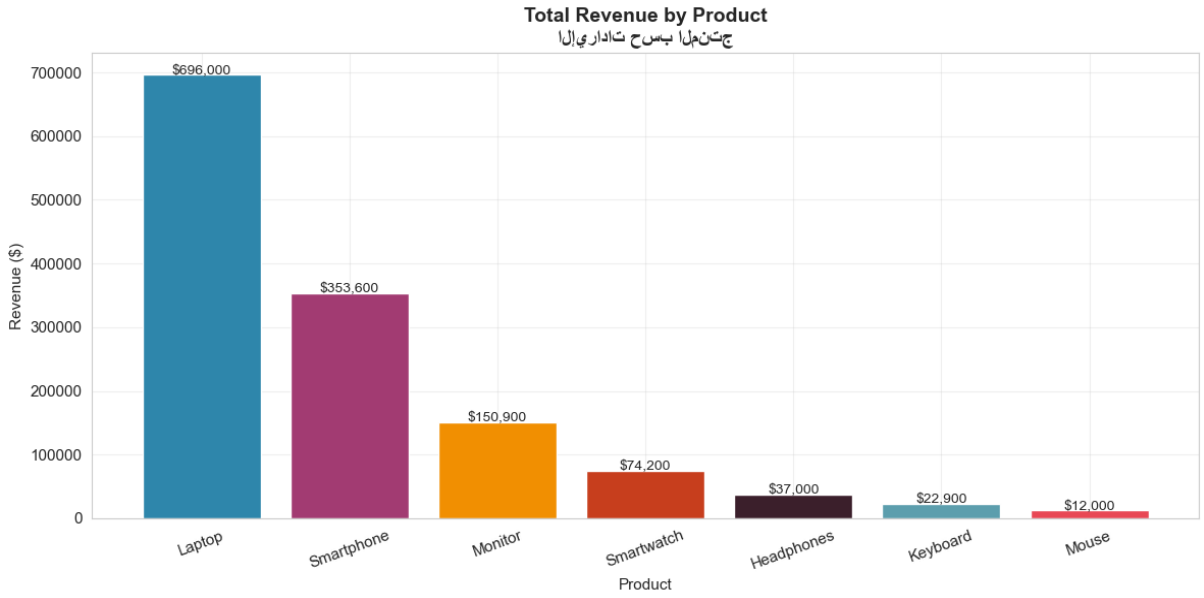
[illegible] [illegible] [illegible]




[illegible] [illegible] [illegible] [illegible] [illegible]. Electronics [illegible] [illegible].

## Revenue by Product

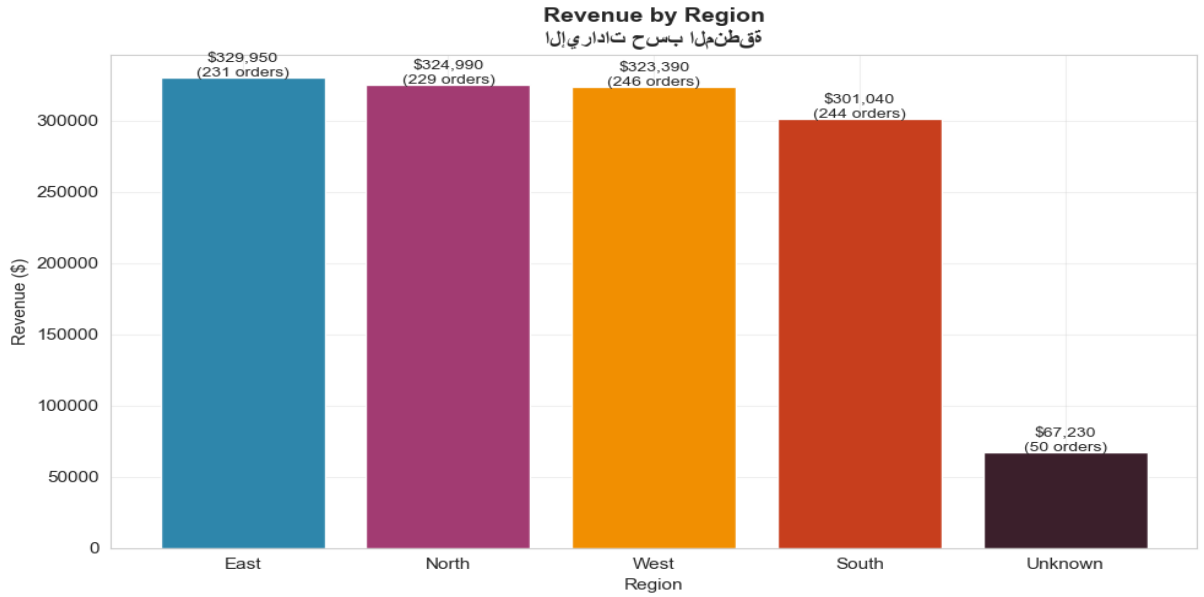
[illegible] [illegible] [illegible]




[illegible] [illegible] [illegible] [illegible]. Laptop [illegible] [illegible].

## Revenue by Region

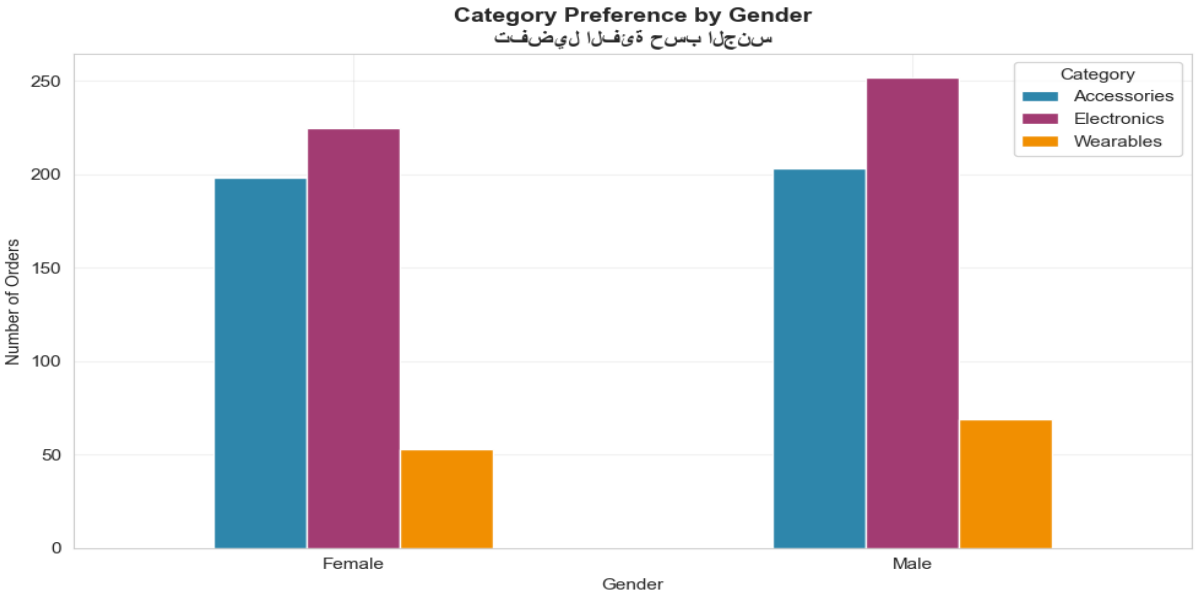
[illegible] [illegible] [illegible]




[illegible] [illegible] [illegible] [illegible]. East [illegible] [illegible].

## Gender vs Category

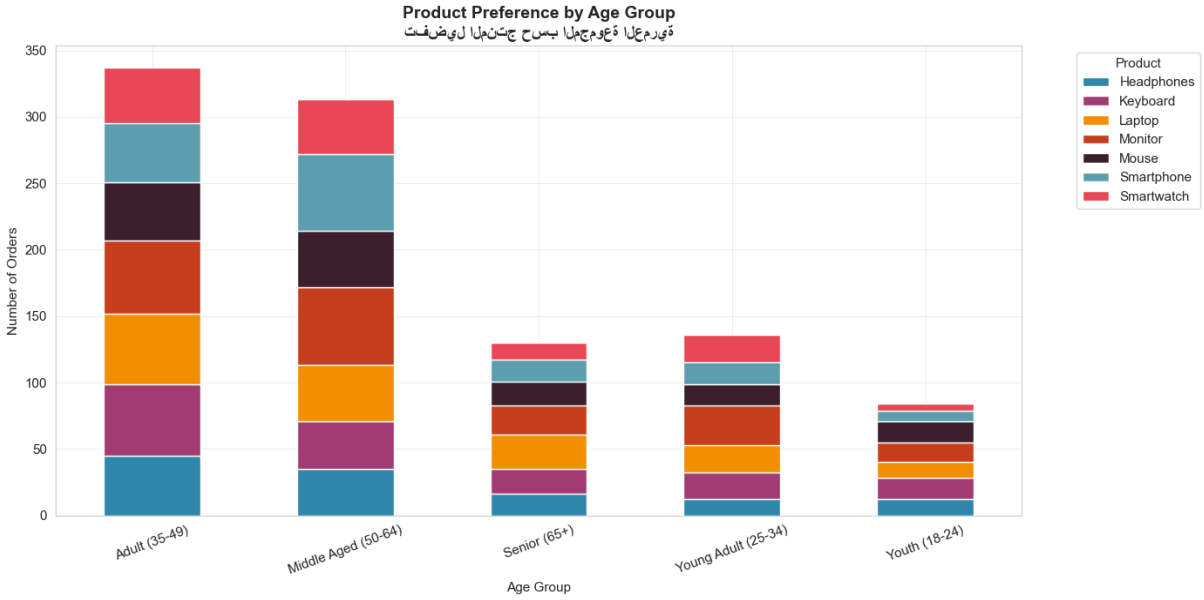
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## Age Group vs Product

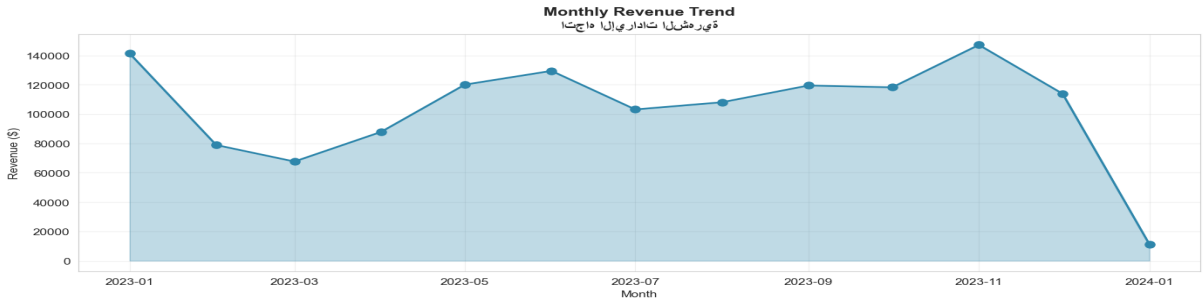
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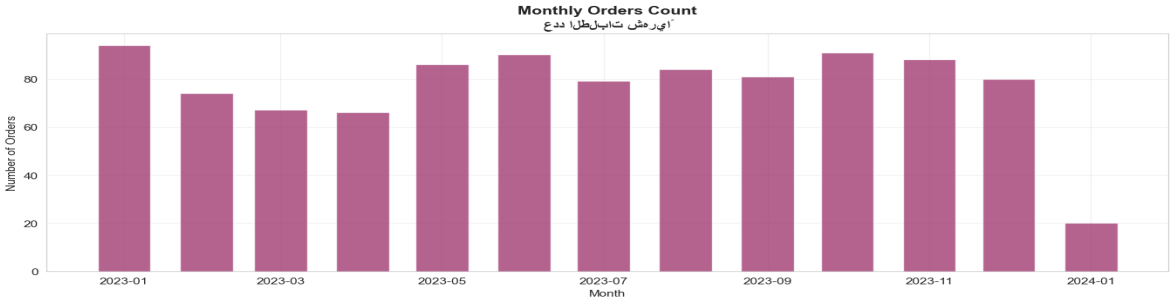



[illegible].

## Monthly Trends

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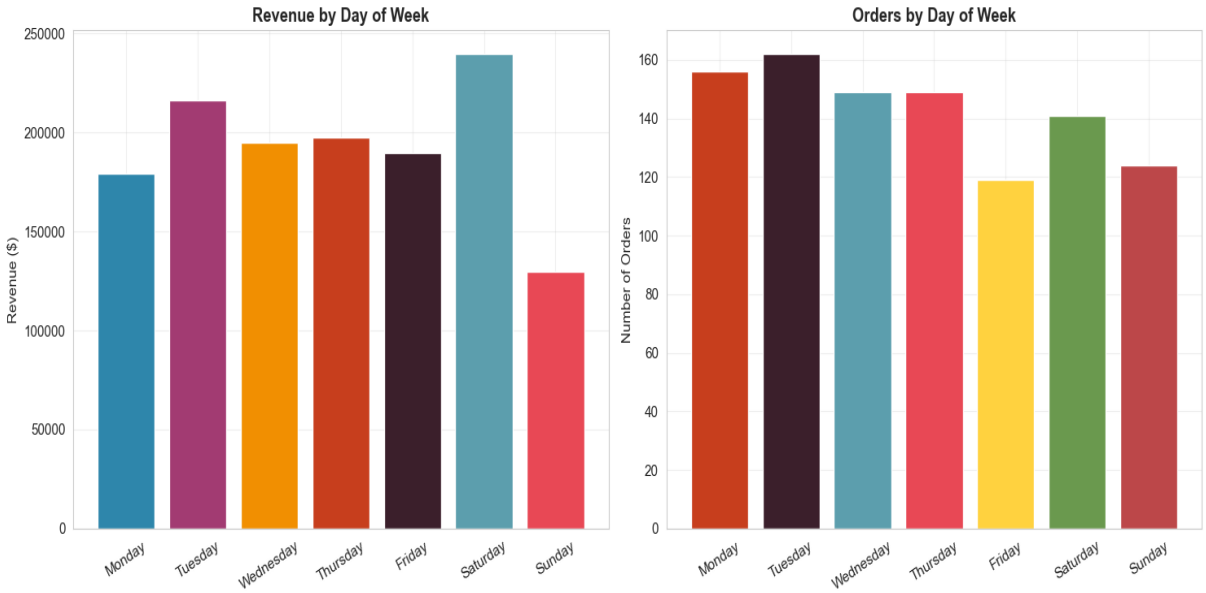



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## Day of Week Analysis

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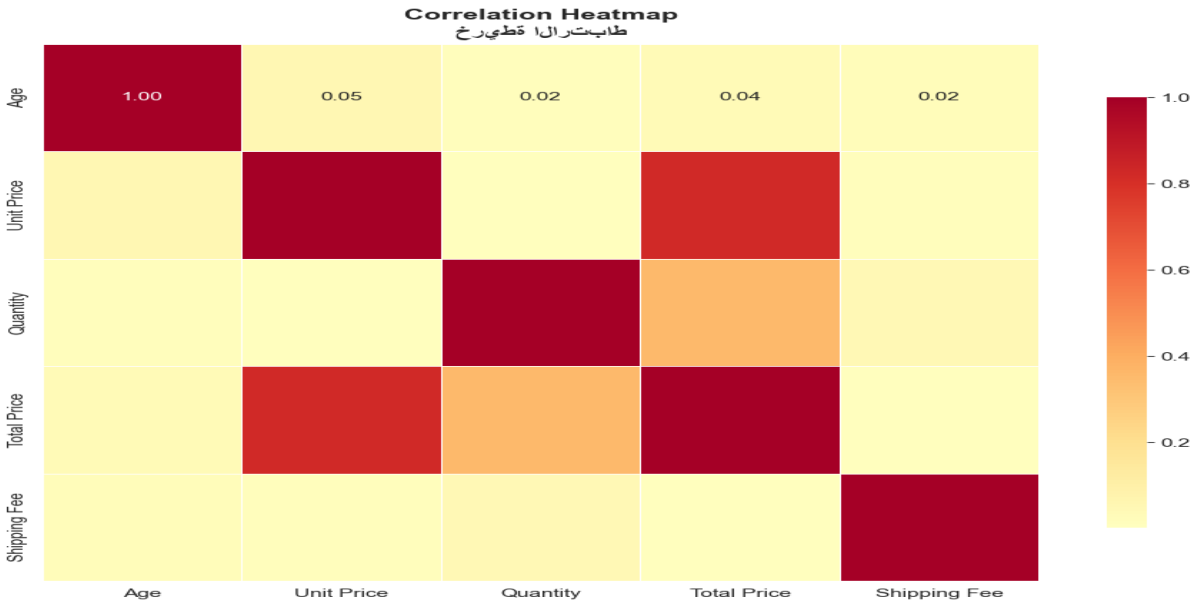




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## Correlation Heatmap

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## Region vs Shipping Status

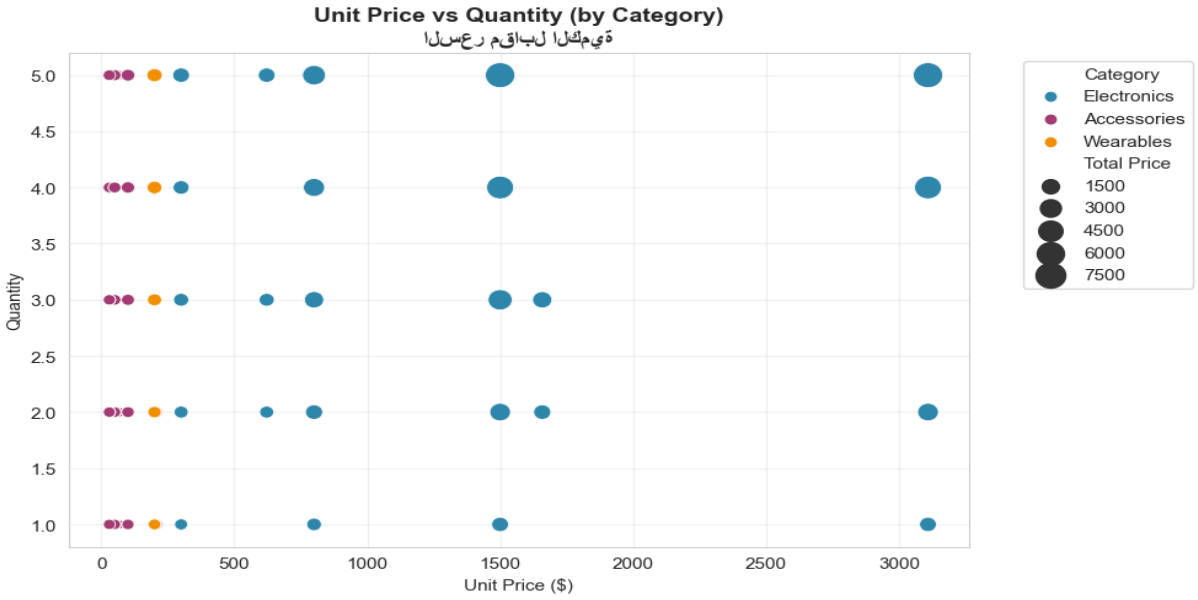
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## Price vs Quantity

[illegible]


Unit Price vs Quantity (by Category)
السعر مقابل الكمية
Quantity
Unit Price ($)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0
500
1000
1500
2000
2500
3000
Category
Electronics
Accessories
Wearables
Total Price
1500
3000
4500
6000
7500

[redacted].

# 4. Key Insights

The following key insights were derived from the analysis:

| # | Insight |
|---|---|
| 1 | ■■■■■■ ■■■■■■■■■■: $1,346,600.00 |
| 2 | ■■■ ■■■■■■■■: 1,000 |
| 3 | ■■■■■ ■■■■■ ■■■■■■: $1346.60 |
| 4 | ■■■■ ■■■■: Electronics ($1,200,500.00) |
| 5 | ■■■■ ■■■■■: Laptop ($696,000.00) |
| 6 | ■■■■ ■■■■■■: East ($329,950.00) |
| 7 | ■■■■■■ ■■■■ ■■■■■■■: 46.9 ■■■■ |
| 8 | ■■■■■■ ■■■■■■■ ■■■■■■: Male (52.4%) |
| 9 | ■■■■■ ■■■■■■■■■■: 30.8% |
| 10 | ■■■■■ ■■■■■■■■ ■■■■■■■■■: 31.3% |
| 11 | ■■■■■■ ■■■■■ ■■■■■■: $12.42 |
| 12 | ■■■■■ ■■■■ ■■■■■■■: November 2023 |
| 13 | ■■■■■ ■■■■ ■■■■■■■: Tuesday (162 ■■■■) |
| 14 | ■■■■■■ ■■■■■■■ ■■■■ ■■■■: 3.01 |

# 5. Recommendations

## 1. Reduce Return Rate

The return rate of 30.4% is significantly high. Investigate product quality issues and improve customer service.

## 2. Focus on Electronics

Electronics generates the highest revenue ($1,179,600). Expand this category.

## 3. Target East Region

East region generates the highest revenue. Develop targeted marketing campaigns.

## 4. November Sales Strategy

November was the peak sales month. Prepare inventory for November each year.

## 5. Tuesday Promotions

Tuesday has the highest orders. Schedule promotions on Tuesdays.

## 6. Age-Based Marketing

Different age groups prefer different products. Develop targeted campaigns.

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**End of Report**